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Welcome

✔ **Video:** Welcome to the Course
1 min

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Experiments and Probability

Probability and Variables

Probability Mass Functions, Probability Density Functions, and Cumulative Distribution Functions

Expected Values, Variances, and Standard Deviation

Random Vectors, Independent Events and Variables, Covariance and Correlation, and Sample Variance

Review

Syllabus

Course Description

Statistics is a thriving discipline that provides the fundamental language of all empirical research. Biostatistics is simply the field of statistics applied in the biomedical sciences.

This course puts forward key mathematical and statistical topics to help students understand biostatistics at a deeper level. After completing this course, students will have a basic level of understanding of the goals, assumptions, benefits and negatives of probability modeling in the medical sciences. This understanding will be invaluable when approaching new statistical topics and will provide students with a framework and foundation for future self learning.

Topics include probability, random variables, distributions, expectations, variances, independence, conditional probabilities, likelihood and some basic inferences based on confidence intervals.

Book

There is an unofficial book for the course, which can be found here: <https://leanpub.com/biostatmethods> ↗ The book was led by Ciprian Crainiceanu, myself and John Muschelli after Ciprian took over the onsite course that this course was based off of. The book is not required, but does make for a nice companion to the course.

Learning Objectives

The goal of this course is to equip biostatistics and quantitative scientists with core applied statistical concepts and methods:

- Students will learn basic mathematical biostatistics including probability distributions and their properties.
- Students will learn the basics of statistical likelihood.
- Students will learn the basics of confidence intervals.
- The course will introduce students to the display and communication of statistical data. This will include graphical and exploratory data analysis using tools like scatterplots, boxplots and the display of multivariate data.

Grading Policy

- Grades are based on student performance on the 4 Module Quizzes. To pass the course, you must earn an 80% or better on each of the four graded quizzes. Students who pass the course and verify their identity will earn a course certificate.
- For calculation of your final grade, the quizzes are weighted as follows: Module 1 Quiz, 38%; Module 2 Quiz, 31%; Module 3 Quiz, 21%; Module 4 Quiz, 10%
- Module homework assignments are for practice only. Although they are graded, your homework grades will have no impact your final course grade. You are encouraged to complete the homework assignments because they will help you master the material covered within this course.

Discussion Forums

All students are encouraged to participate in the Discussion Forums by asking and answering questions and providing feedback.

All students are expected to be civil when participating in the Discussion Forums.

- Be friendly and considerate when talking to your fellow students.
- Use up-votes to bring attention to thoughtful, helpful posts.
- Search before you post.
- Post in the appropriate sub-forum.
- Use the flag icon to report inappropriate content or highlight posts related to platform issues.

Mark as completed

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